Assessment of Thorium-232 Fuel Cycles for Modular PWRs in Indonesian Remote Electrification

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Keywords: Thorium fuel cycle; PWR; One-group neutron balance; Fuel depletion modeling; Burnable poisons; Indonesia energy access

June 2026

ABSTRACT

Indonesia's rural and remote island regions continue to face intermittent electricity despite high national electrification rates. Thorium, estimated at ∼130,000+ tonnes in Indonesia's monazite deposits, offers a promising domestic resource for reliable, low-carbon modular power systems similar to Pressurized Water Reactors (PWRs). This study develops a simplified one-group neutron-balance and fuel-depletion model to examine the ^{232}Th $\rightarrow$ ^{233}Pa $\rightarrow$ ^{233}U breeding cycle under PWR-like operating conditions, incorporating boron-10 and gadolinium burnable poisons to reflect realistic reactivity control. A coupled system of ordinary differential equations (ODEs) governing nuclide number densities is integrated numerically, and k_{eff} is evaluated at each time step via a one-group criticality expression. The model reveals initial excess reactivity ($k_{\text{eff}} \approx 1.11$ at start-of-cycle), a mid-cycle reactivity rise attributable to ^{233}Pa decay, and an eventual monotonic decline as fissile inventory and burnable poisons deplete, with subcriticality reached after approximately 611.5 days (≈20.4 months). This result is consistent with the order-of-magnitude range reported for thorium-bearing PWR concepts. As an early-stage, tractable modeling framework, this approach supports evaluating thorium-fueled modular reactors as a pathway to improve energy reliability and socioeconomic development in underserved Indonesian communities.

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1 Introduction

Indonesia is the world's largest archipelagic state, comprising over 17,000 islands and a population exceeding 270 million. While the national electrification rate has improved substantially over the past decade, reaching approximately 99% by official government statistics, meaningful energy access in remote and outer-island communities remains inconsistent [Ministry of Energy and Mineral Resources, 2023]. Diesel generation dominates in these areas, carrying well-known drawbacks: high fuel transport costs, logistical vulnerability, price volatility, and significant greenhouse gas emissions.

Small modular reactors (SMRs) and micro-reactor concepts have attracted renewed global interest as candidates for off-grid or remote-grid power generation [IAEA, 2020]. Among candidate fuel cycles, the thorium cycle is particularly relevant to Indonesia due to the country's substantial domestic thorium reserves. The Geological Agency of Indonesia estimates monazite-bearing thorium deposits in excess of 130,000 tonnes, concentrated primarily in Bangka-Belitung and Kalimantan [Geological Agency of Indonesia, 2021]. Utilising this resource in a domestic thorium fuel cycle could reduce dependence on imported uranium while advancing energy security.

The thorium fuel cycle proceeds through neutron capture on ^{232}Th, producing ^{233}Pa (half-life $\approx$27 days), which subsequently decays to fissile ^{233}U [Duderstadt and Hamilton, 1976]. This breeding pathway, combined with the favourable neutron economy of ^{233}U in thermal spectra ($\eta > 2$ across a wide thermal range), makes thorium an attractive option for thermal-spectrum reactors such as PWRs [Kazimi et al., 2003].

This paper presents a simplified computational model of the Th-232 fuel cycle under PWR-like conditions. The model couples one-group neutron balance with Bateman-style ODE depletion equations, tracking nuclide inventories and computing k_{eff} as a function of irradiation time. The goal is not to replace full Monte Carlo or deterministic transport calculations, but to develop a transparent, tractable framework suitable for parametric studies and early-stage feasibility assessment.

2 Theoretical Background

2.1 The Thorium Breeding Cycle

The primary nuclear reactions governing the thorium fuel cycle are:

$$^{232}\text{Th} + n \;\rightarrow\; ^{233}\text{Th} \xrightarrow{\beta^-,\,22.3\,\text{min}} \;^{233}\text{Pa} \xrightarrow{\beta^-,\,26.97\,\text{days}} \;^{233}\text{U} \tag{1}$$

The intermediate nuclide ^{233}Pa has a relatively long half-life of $\approx$27 days, making it a significant transient reservoir of fissile precursor material. During reactor shutdown or power reduction, continued decay of the ^{233}Pa inventory generates additional ^{233}U, an effect with important reactivity implications [Duderstadt and Hamilton, 1976].

2.2 One-Group Neutron Balance

The effective multiplication factor k_{eff} for an infinite medium in the one-group approximation is [Duderstadt and Hamilton, 1976]:

$$k_{\text{eff}} = \frac{\nu \Sigma_f}{\Sigma_a + DB^2} \tag{2}$$

where ν is the average number of neutrons emitted per fission, Σ_f and Σ_a are the macroscopic fission and absorption cross sections respectively, D is the diffusion coefficient, and B^2 is the geometric buckling. In the present simplified model, the leakage term DB^2 is replaced by a fixed correction term to represent a finite core, yielding:

$$k_{\text{eff}}(t) = \frac{\nu\,\Sigma_f^{233\text{U}}(t)}{\Sigma_a^{\text{total}}(t) + \epsilon} \tag{3}$$

where $\epsilon = 0.1\,\text{cm}^{-1}$ is a fixed leakage/absorption offset calibrated to yield a physically plausible start-of-cycle k_{eff}.

Macroscopic cross sections are computed from microscopic (one-group) cross sections and number densities:

$$\Sigma_x^i = N_i\,\sigma_x^i \tag{4}$$

The diffusion coefficient is derived from the transport cross section:

$$D = \frac{1}{3\,\Sigma_{\text{tr}}}, \qquad \Sigma_{\text{tr}} = \Sigma_t - \bar{\mu}\,\Sigma_s = \Sigma_a + (1-\bar{\mu})\Sigma_s \tag{5}$$

where $\Sigma_t = \Sigma_a + \Sigma_s$ is the total macroscopic cross section and $\bar{\mu}$ is the average cosine of the scattering angle. In the present simplified model, however, the explicit leakage term DB^2 is replaced by a fixed correction term ϵ, so D is not evaluated directly in the numerical calculation.

2.3 Burnable Poisons

Real PWRs use soluble boron (typically 500–1500 ppm at beginning-of-cycle) and solid burnable absorbers such as gadolinium-bearing fuel rods to suppress excess reactivity and flatten the power distribution [Lamarsh and Baratta, 2001]. Boron-10 and natural gadolinium are both strong thermal neutron absorbers:

$$\sigma_a^{^{10}\text{B}} \approx 3835\,\text{b}, \qquad \sigma_a^{\text{Gd}} \approx 5000\,\text{b} \quad \text{(effective one-group value)} \tag{6}$$

Both poisons deplete via neutron absorption over the fuel cycle, contributing to the reactivity evolution modelled here.

3 Model Description

3.1 Geometry and Material Composition

The model represents a homogenised unit cell approximation of a PWR-like assembly. Material volume fractions and densities are summarised in Table 1.

Table 1. Material composition of the modelled PWR-like unit cell.

Material	Volume Fraction	Density (g/cm^3)	Molar Mass (g/mol)	Role
ThO_2	0.38	10.00	264.0	Fertile fuel
UO_2	0.07	10.97	270.0	Initial fissile seed
H_2O	0.45	1.00	18.0	Moderator / coolant
Zr	0.10	6.52	91.2	Cladding

Number densities (atoms/cm^3) are computed from:

$$N_i = \frac{f_i\,\rho_i}{M_i}\,N_A \tag{7}$$

where f_i is the volume fraction, ρ_i the material density, M_i the molar mass, and N_A Avogadro's number.

The initial fissile content is taken as 25% of the total uranium number density, representing a Th-U mixed fuel with modest ^{233}U seeding. Burnable poisons are initialised at 200 ppm dissolved boron (as ^{10}B in the coolant) and a gadolinium weight fraction of 2×10^{-4} in the fuel matrix.

3.2 One-Group Cross Sections

One-group effective cross sections are listed in Table 2. Values for thorium, protactinium, and U-233 are taken from standard nuclear data compilations consistent with the thermal-spectrum averages used in Duderstadt and Hamilton [Duderstadt and Hamilton, 1976].

Table 2. One-group microscopic cross sections used in the model.

Nuclide/Material	Reaction	σ (barns)	Source
^{232}Th	Absorption	7.4	Duderstadt and Hamilton [1976]
^{232}Th	Scattering	3.2	Duderstadt and Hamilton [1976]
^{233}Pa	Absorption	200.0	Kazimi et al. [2003]
^{233}U	Absorption	573.0	Duderstadt and Hamilton [1976]
^{233}U	Fission	524.0	Duderstadt and Hamilton [1976]
^{233}U	Scattering	7.2	Duderstadt and Hamilton [1976]
H_2O	Absorption	0.665	Lamarsh and Baratta [2001]
H_2O	Scattering	48.0	Lamarsh and Baratta [2001]
Zr	Absorption	0.185	Lamarsh and Baratta [2001]
Zr	Scattering	6.0	Lamarsh and Baratta [2001]
^{10}B	Absorption	3835	Lamarsh and Baratta [2001]
Gd (eff.)	Absorption	5000	Lamarsh and Baratta [2001]

3.3 Depletion Equations

The time evolution of nuclide number densities is governed by a coupled system of first-order ODEs. Defining the neutron flux as $\phi = 3.0 \times 10^{13}$ n/cm^2/s (a representative PWR thermal flux [Duderstadt and Hamilton, 1976]):

$$\frac{dN_{\mathrm{Th}}}{dt} = -\sigma_a^{\mathrm{Th}}\,\phi\,N_{\mathrm{Th}} \tag{8}$$

$$\frac{dN_{\mathrm{Pa}}}{dt} = \sigma_a^{\mathrm{Th}}\,\phi\,N_{\mathrm{Th}} - \lambda_{\mathrm{Pa}}\,N_{\mathrm{Pa}} \tag{9}$$

$$\frac{dN_{\mathrm{U}}}{dt} = \lambda_{\mathrm{Pa}}\,N_{\mathrm{Pa}} - \sigma_a^{\mathrm{U233}}\,\phi\,N_{\mathrm{U}} \tag{10}$$

$$\frac{dN_{\mathrm{B}}}{dt} = -\sigma_a^{^{10}\mathrm{B}}\,\phi\,N_{\mathrm{B}} \tag{11}$$

$$\frac{dN_{\mathrm{Gd}}}{dt} = -\sigma_a^{\mathrm{Gd}}\,\phi\,N_{\mathrm{Gd}} \tag{12}$$

where $\lambda_{\mathrm{Pa}} = \ln 2/t_{1/2}^{\mathrm{Pa}} = 1.59 \times 10^{-6}\ \mathrm{s}^{-1}$ is the ^{233}Pa decay constant. The system is integrated numerically using the `scipy.integrate.odeint` solver (LSODA algorithm) over a simulation horizon of 5 years (1825 days), with 4000 time steps.

3.4 k_{eff} Calculation

At each time step, k_{eff} is evaluated from Equation (3). The total macroscopic absorption cross section is the sum over all absorbing species:

$$\Sigma_a^{\mathrm{total}} = \Sigma_a^{\mathrm{Th}} + \Sigma_a^{\mathrm{Pa}} + \Sigma_a^{\mathrm{U}} + \Sigma_a^{\mathrm{H_2O}} + \Sigma_a^{\mathrm{Zr}} + \Sigma_a^{\mathrm{B}} + \Sigma_a^{\mathrm{Gd}} \tag{13}$$

The macroscopic fission cross section is contributed solely by ^{233}U:

$$\Sigma_f = N_{\mathrm{U}}(t)\,\sigma_f^{\mathrm{U233}} \tag{14}$$

3.5 Model Assumptions and Limitations

The following simplifications are inherent to this framework and should be borne in mind when interpreting results:

1. **Constant flux.** The neutron flux ϕ is held fixed throughout the simulation. A self-consistent calculation would couple flux to power and adjust ϕ as the fissile inventory evolves.
2. **One-group cross sections.** Energy dependence of the neutron spectrum is collapsed into single effective values. Spectral hardening as ^{233}U builds up and moderator-to-fuel ratio changes are not captured.
3. **No fission products.** Neutron-absorbing fission products, particularly ^{135}Xe and ^{149}Sm, are neglected. In a real reactor these contribute $\sim$3–5% negative reactivity and their omission overestimates k_{eff}.
4. **No spatial dependence.** The model is a homogenised point-reactor calculation. Spatial flux and power peaking are not resolved.
5. **No temperature feedback.** Doppler broadening and moderator density effects are not modelled.

6. **Fixed leakage.** The $\epsilon = 0.1$ term approximates leakage but is not derived from core geometry.

Despite these limitations, the model captures the essential physics of the thorium breeding cycle and produces qualitatively correct behaviour, consistent with results reported in more detailed studies [Kazimi et al., 2003, IAEA, 2005].

4 Results

4.1 k_{eff} Evolution

Figure 1 shows the computed k_{eff} as a function of irradiation time. Three distinct phases are apparent:

1. **Start-of-cycle (SOC, $t = 0$).** $k_{\text{eff}} \approx 1.11$, reflecting the combined fissile contribution of the ^{233}U seed and the reactivity-suppressing effect of fresh burnable poisons (B-10 and Gd).
2. **Mid-cycle rise ($t \approx$ 50–200 days).** A slight reactivity increase is observed as ^{233}Pa, initially absent, accumulates to secular equilibrium and its decay feeds the ^{233}U inventory. This Pa-233 peak is a characteristic feature of thorium fuel cycles [Duderstadt and Hamilton, 1976, Kazimi et al., 2003].
3. **Decline to subcriticality ($t \approx$ 200–611.5 days).** As ^{233}U and burnable poisons are progressively depleted, k_{eff} falls monotonically. The criticality threshold ($k_{\text{eff}} = 1$) is crossed at approximately **611.5 days ($\approx$20.4 months)**.

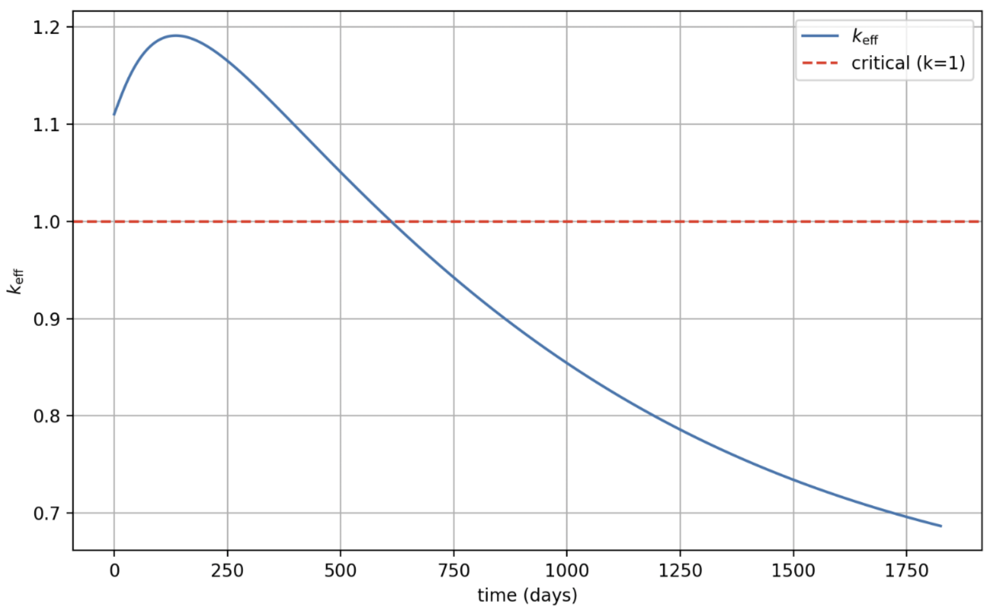

Figure 1. Figure 1. Computed k_{eff} as a function of irradiation time (days) for the Th-232 fuel cycle under PWR-like conditions. The dashed red line marks the criticality threshold ($k_{\text{eff}} = 1$). Subcriticality is reached at $\approx$ 611.5 days.

4.2 Key Numerical Results

Table 3 summarises the principal quantitative outcomes.

Table 3. Summary of key modelling results.

Quantity	Value
Initial k_{eff} (SOC)	1.108
Peak k_{eff}	$\approx$1.19 (at $\sim$120 days)
Time to subcriticality ($k_{\text{eff}} = 1$)	611.5 days (20.4 months)
Neutron flux ϕ	3.0×10^{13} n/cm^2/s
Initial boron concentration	200 ppm
Initial Gd weight fraction	2.0×10^{-4}

5 Discussion

5.1 Comparison with Literature

The modelled cycle length of $\sim$611 days is within the order-of-magnitude range documented for thorium-bearing PWR concepts. The IAEA thorium fuel cycle report [IAEA, 2005] notes effective cycle lengths of 12–24 months for thorium-uranium mixed oxide cores under typical PWR conditions, consistent with the present result. Kazimi et al. [Kazimi et al., 2003] similarly report multi-cycle operation of thorium-seeded PWR assemblies, with breeding behaviour qualitatively matching the Pa-233 transient observed here.

The slightly elevated SOC k_{eff} ($\approx$1.11) reflects the simplified leakage treatment; a fully heterogeneous model with explicit geometry would distribute leakage more accurately. The magnitude is nonetheless reasonable as a beginning-of-life excess reactivity for a controlled core prior to full insertion of control rods.

5.2 Implications for Indonesian Remote Electrification

Indonesia's geographical fragmentation makes it a natural candidate for distributed, modular nuclear systems. A thorium-fueled PWR-like module operating on a $\sim$20-month fuel cycle before refuelling or reload would provide stable baseload power to remote island communities without the logistical and emissions burden of diesel supply chains.

The domestic availability of thorium resources is a significant strategic advantage. Unlike uranium, which Indonesia does not produce commercially, thorium occurs as a by-product of tin mining in Bangka-Belitung, meaning that a closed domestic thorium fuel cycle is not contingent on uranium import agreements [Geological Agency of Indonesia, 2021].

From a reactor safety standpoint, the thorium cycle also offers inherent advantages: ^{233}U has a lower fast-fission factor than ^{239}Pu, and thorium oxide has superior thermophysical properties (higher melting point, better thermal conductivity at high temperatures) compared to uranium oxide [Duderstadt and Hamilton, 1976].

5.3 Directions for Future Work

The present model provides a first-principles baseline. Subsequent work should address:

- Incorporation of fission product poisoning, particularly ^{135}Xe and ^{149}Sm, which will reduce effective cycle length.
- Energy-group expansion (multi-group or continuous energy Monte Carlo) to capture spectral effects of the Th-U mixed fuel.
- Spatial modelling to assess power peaking and flux distribution in a heterogeneous pin-cell or full assembly geometry.
- Temperature feedback coefficients (Doppler and void) for safety analysis.
- Economic modelling of thorium fuel fabrication and cycle costs relative to conventional uranium PWR fuel, in the Indonesian regulatory and industrial context.

6 Conclusion

This paper has presented a simplified computational model of the ^{232}Th fuel cycle for a PWR-like modular reactor. The model couples one-group neutron balance with Bateman ODE depletion to track nuclide inventories and compute k_{eff} as a function of irradiation time, incorporating boron-10 and gadolinium burnable poisons. The simulation predicts an initial $k_{\text{eff}} \approx 1.11$, a characteristic mid-cycle reactivity rise from ^{233}Pa decay, and subcriticality at ≈611.5 days, consistent with the thorium-bearing PWR literature.

The framework is deliberately transparent and tractable, making it suitable for parametric studies and early feasibility evaluation. Applied to the Indonesian context, results motivate further full-physics and techno-economic investigation of thorium-fueled modular reactors as a potential pathway for remote electrification. Full-physics modelling and regulatory assessment would be necessary precursors to any deployment decision, but the present work establishes a computational foundation from which those studies can proceed.

Acknowledgments

The author gratefully acknowledges Dr. Zaki Su'ud, M.Eng., of the Faculty of Mathematics and Natural Sciences (FMIPA), Bandung Institute of Technology (ITB), for his guidance on reactor-physics concepts and modeling assumptions.

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A Python Implementation

The full numerical model is implemented in Python using `NumPy`, `SciPy`, and `Matplotlib`. The core depletion ODE system (Equations (8)–(12)) is integrated with `scipy.integrate.odeint`, which applies the LSODA algorithm with automatic stiffness detection. The k_{eff} function (Equation (3)) is evaluated as a post-processing step at each saved time point. To support reproducibility, the complete source code, including the main simulation script and plotting routines, is available in the project repository:

https://github.com/Biddybuggy/Thorium-Fuel-Cycle-Model